Preliminary Examination of Heterogeneous Effects of Free Trade Agreements*

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Abstract

Using a random coefficients empirical model, we show that economic integration agreements (EIAs) have heterogenous effects on trade flows. Furthermore, we identify that the effects of an EIA on trade creation are related to the causes of an EIA existing.

Key words: International trade, economic integration agreements

JEL classification: F1, F21, F22, F23

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One of the most dramatic changes to the international landscape over the last thirty years has been the growth of economic integration agreements (EIAs). Article XXIV of the General Agreement on Tariffs and Trade (GATT) established a means for countries to negotiate trade agreements outside of the GATT (and World Trade Organization, or WTO). Initially, these agreements were small in number and in scope. However, over the last thirty years the admissible agreements under Article XXIV have grown in number and in scope. Baier and Bergstrand (2007) report that in 1965 there were only 115 trade agreements in existence. By 1980 the number of EIAs had increased to 499 and by 2005 there were 1760 such agreements. These agreements have grown not only in number but also in scope. Most agreements in 1965 were free trade agreements that lowered tariffs and non-tariff barriers. Over time these and subsequent agreements evolved into customs unions (with common external tariffs), common markets (with factor mobility) and economic unions (with factor mobility, common external tariffs and coordinated monetary and/or fiscal policies). The growth in the number of EIAs motivated many scholars in economics and political science to try to better understand the causes and the consequences of increased EIAs.

In terms of consequences, much of the initial academic research focused on estimating how much trade is created (or diverted) by these agreements. These early estimates of the trade-creating effects of EIAs produced mixed results. Recent developments in economic theory and econometric techniques have produced results that indicate that EIAs increase trade considerably, cf., Baier and Bergstrand (2007). The early studies estimates were likely biased because important variables were omitted – for example, most studies did not control for multilateral resistance terms – and most studies did

not control for the potential endogeneity of the EIAs. In correcting for the omitted variables and the potential endogeneity, it was common for researchers to treat all trade agreements' effects as the same. While most scholars believed there was likely to be a great deal of heterogeneity across agreements, the analysis focused on measuring an average treatment effect and assuming the heterogeneity was uncorrelated with other variables in the analysis and unobserved by the researcher.

At the same time, economists and political scientists were also providing theoretical and empirical insights into the factors that *caused*, or at least were correlated with, EIAs. In this literature, economists typically focused on economic factors that increased the likelihood of economic integration via increased trade flows, cf., Baier and Bergstrand (2004). Political scientists typically focused on political economy aspects and institutional features of the countries that may influence the likelihood of trade agreements, cf., Mansfield and Milner (1999). In terms of the determinants of trade flows, researchers have identified factors that are associated with increased trade and institutional factors that may be associated with the costs of negotiating the trade agreements.

This objective of this paper is to show how these literatures complement each another. We use a *random coefficient model*, which allows for heterogenous effects of the EIAs. We show that the *consequences* (trade-creating effects) of an EIA are associated with the economic and political *causes* of the agreement. The analysis suggests that more can be learned about the trade-creating effects of the trade agreements by moving away from a one-size-fits-all model, and the insights in this paper may provide more precise *ex ante* predictions of the trade-creating effects.

I. Econometric Issues

For near a half-century the gravity equation has been the workhorse for empirical analyses of international trade flows and quantifying the effects of *economic integration agreements* as well as other trade costs. Tinbergen (1962) was the first cross-sectional empirical study of trade flows using the the gravity framework to quantify the trade creating effects of membership in the British Commonwealth or the Benelux agreement. Using dummy variables for country-pairs in an agreement, Tinbergen found that membership in the British Commonwealth (Benelux) increased (bilateral) trade flows by five (four) percent. Subsequent cross-sectional empirical investigations provided mixed support for the trade creating effects of *economic integration agreements.* For example, Aitken (1973), Abrams (1980) and Brada and Mendez (1985) found that economic integration agreements tended to increase trade. However, Bergstrand (1985), Frankel, Stein, and Wei (1995), and Frankel (1997) found that some EIAs raised trade, others had no economically or statistically significant effect, and some EIA's apparently lowered trade. Frankel (1997) found that membership in the European Union is often associated with *lower* bilateral trade. This is surprising since the European Union is one of the oldest agreements that has also grown in membership and depth of integration over time.

Using a series of cross-sectional specifications, Baier and Bergstrand (2002) show that, among all the standard gravity covariates, the dummy variable coefficient estimate for free trade agreements is the least stable. Baier and Bergstrand (2007) show that the instability is likely attributable endogeneity bias; they argue that the endogeneity

bias arises from the fact that free trade agreements are not randomly assigned and that in order to obtain consistent estimates of the impact of trade agreements on trade flows it is important control for their potential endogeneity. In order to address the endogeneity of the agreements, standard econometric techniques provides two paths. An instrumental variable approach can be applied if there exists a set of variables that are correlated with EIA membership, but are uncorrelated with bilateral trade flows. Alternatively, if the (bilateral) factors that lead to the endogeneity are *fixed*, or slow moving, over time, then panel data techniques (fixed-effects or first-differencing) can produce consistent estimates of the trade-creating impacts of trade agreements on bilateral trade flows. Using the latter approach, Baier and Bergstrand (2007) find that the (partial equilibrium) average treatment effect of a free trade agreement results in a doubling of bilateral trade. More recently, Anderson and Yotov (2011), using a similar technique with more disaggregated trade data, find results that are quantitatively similar to Baier and Bergstrand. Egger, Larch, Staub, and Winkelmann (2011) using an IV approach find that trade agreements tend to raise trade as well. Thus, it seems clear that – once one accounts for the potential endogeneity of EIAs – they tend to raise bilateral trade flows.

While correcting for endogeneity results in consistent estimates of the average treatment effect of an EIA, it seems unlikely that the average treatment effect is the same for all types of agreements and country-pairs. In order to assess the impact of different trade agreements on bilateral trade flows, it is possible to include a full set of dummy variables for each agreement. This type of analyses was carried out by Baier, Bergstrand, and Vidal (2008) and Baier, Bergstrand, Egger, and McLaugh-

lin (2008), Egger and Larch (2011), and Kohl (2013). Similar, to the earlier cross sectional analyses the trade-creating impacts of the agreements in these papers are mixed. There are several factors that limit the applicability of this framework. First by using a full set of dummy variables, it seems likely that an IV approach is under-identified; that is, there are more endogenous variables than instruments. Therefore, most authors have employed panel data methods to assess the trade creating effects of specific EIAs. There are, however, several issues with this type of analysis. Once the researcher controls for exporter-time, importer-time, and bilateral country-pair fixed effects, statistical tests may be weak for some agreements because the agreement may be small in scope (limiting the number of pair) and/or in terms of duration; in either case, the limited variation in the RHS dummy may result in a weak estimated effect. Second, for some country pairs, it may be the case that extrapolation from the OLS estimates may not provide the right counterfactual (cf., Baier and Bergstrand (2009)) for a more thorough discussion on this issue. Finally, these studies are more suited for *ex post* analysis, and provide little guidance for *ex ante* policy analyses. Taking a different approach, Baier, Bergstrand and Feng (2013) use a set dummy variables that categorizes the depth of the EIA. In their paper, they find that the deeper integration tends to create more trade; that is, the average treatment effect tends to be higher if the bilateral pair is in a customs union, common market, or economics union rather than a free trade agreement. In this paper, we allow for heterogeneity in EIAs' effects that are related to economic and political fundamentals that may influence the nature of the agreement. We show that, by using economic and political factors that are associated with the decision to enter an EIA, we can account for some of

the heterogeneity of the trade-creating effects and that these factors can contribute to our understanding of the evolution of EIAs around the world.

II. Econometric Specification

The earliest applications of the gravity equation to international trade flows were not based on formal economic models; c.f. Tinbergen (1962), Linnemann (1966), Aitken (1973), and Sapir (1981). Over the last 30 years, several authors have shown that a gravity structure emerges under a variety of different underlying assumptions related to technology, preferences, and market structure, c.f. Anderson(1979), Bergstrand (1985), Deardorff (1998), Eaton and Kortum (2001), Helpman, Melitz, and Rubinstein (2008), Bernard, Jensen, Redding, and Schott (2012), and Novy (2013). The empirical gravity specification typically includes time-invariant and time-varying (proxy) measures of bilateral trade costs, time-varying source (destination) effects, and policy variables that may influence trade flows. Baier and Bergstrand (2007) show that consistent estimation of the gravity structure is complicated because the policy variables are (potentially) endogenous. To obtain consistent estimates, researchers employ panel data or instrumental variable techniques to account for the potential endogeneity.

In this paper, we follow Baier and Bergstrand (2007) by assuming the bilateral factors that are correlated with the economic integration agreements are constant or slow moving over time. Therefore, we write the gravity equation as

$$TF_{ijt} = exp\Big\{\phi + x_{ijt}^{TC}\beta^{TC} + x_{it}^{S}\beta^{S} + x_{jt}^{D}\beta^{D} + \alpha_{ij}\mathbf{w}_{ijt} + c_{ij} + \nu_{ijt}\Big\} \tag{1}$$

where TF_{ijt} are trade flows from country i to county j at time t, x_{ijt}^{TC} is a vector of bilateral measures of trade costs that impact trade between the country pair at time t, x_{it}^{S} is a vector of variables related to the source country (i) at time t, x_{it}^{D} is a vector of variables related to the destination country (j) at time t, $\mathbf{w}_{ijt}$ is a dummy variable that is equal to unity if the pair of countries has an EIA and zero otherwise, α_{ij} is the bilateral-pair related effect associated with an EIA between countries i and j, c_{ij} is the unobserved bilateral effect that is potentially correlated with the other RHS variables, in particular the economic integration agreement, ν_{ijt} is the unobserved component that is assumed to be uncorrelated with the RHS variables, and *exp* denotes the exponentiated value of the term in brackets. The conditional expectations $(E[\,|\,\cdot])$ is given by

$$E\Big[TF_{ijt}|x,\,\mathbf{w},\,\alpha_{ij}\,,c_{ij}\Big] = exp\Big\{\phi + x_{ijt}^{TC}\beta^{BC} + x_{it}^{S}\beta^{S} + x_{jt}^{D}\beta^{D} + \alpha_{ij}\mathbf{w}_{ijt} + c_{ij}\Big\}$$

where we maintain the assumption strict exogeneity assumption such that $y = (y_{ij1,}y_{ij2}, ..., y_{ijT})$

Ignoring zeros for now, the equation above can be written in log form as (1):

$$ln\Big(TF_{ij}\Big) = \phi + x_{ijt}^{TC}\beta^{BC} + x_{it}^{S}\beta^{S} + x_{jt}^{D}\beta^{D} + \alpha_{ij}\mathbf{w}_{ijt} + c_{ij} + \nu_{ijt}$$

and the conditional expectations of this expression is:

$$E\Big[ln\Big(TF_{ij}\Big)|x,\ \mathbf{w},\ \alpha_{ij}\ , c_{ij}\Big] = \phi + x_{ijt}^{TC}\beta^{BC} + x_{it}^{S}\beta^{S} + x_{jt}^{D}\beta^{D} + \alpha_{ij}\mathbf{w}_{ijt} + c_{ij}$$

To highlight our empirical approach in accounting for random coefficients and to economize on notation, we write the above equation succinctly as:

$$E\Big[ln\Big(TF_{ij}\Big)|x,\ \mathbf{w},\ \alpha_{ij}\ , c_{ij}\Big] = \mu(\mathbf{x}) + \alpha_{ij}\mathbf{w}_{ijt} + c_{ij}$$

where $\mu(\mathbf{x}) = \phi + x_{ijt}^{TC}\beta^{TC} + x_{it}^{S}\beta^{S} + x_{jt}^{D}\beta^{D}$.

The random coefficient model allows the α_{ij} to potentially vary across country pairs. More specifically, we assume that the heterogeneity in the treatment depends on a vector of factors (z) that are observable to the econometrician as well as an unobserved component. We assume that the z may include factors in $\mathbf{x}$. As in Wooldridge (2010), we assume that random coefficient can be written as

$$\alpha_{ij} = \alpha + a_{ij}$$

where the unconditional expectation of the treatment effect is given by $E(\alpha_{ij}) = \alpha$ and $E\Big[\alpha_{ij}|x\ , w\ , c_{ij},\ z\Big] = \alpha + E(a_{ij}|z)$ where $E(a_{ij}|z) \neq 0$. In the specification employed in this paper, we assume that econometric models satisfies the following conditions:

$$E\Big[ln\Big(TF_{ij}\Big)|x,\ \mathbf{w},\ \alpha_{ij},\ c_{ij},\ z\Big] = \mu(\mathbf{x}) + \alpha_{ij}\mathbf{w}_{ijt} + c_{ij} \tag{2}$$

$$E\Big[\alpha_{ij}|x,\ \mathbf{w}\ , c_{ij},\ z\Big] = \alpha + (\bar{z}_{ij} - \mu(z))\alpha_z \tag{3}$$

$$E\Big[ln\Big(TF_{ij}\Big)|x,\ \mathbf{w},\ c_{ij},\ z\Big] = \mu(\mathbf{x}) + \alpha\mathbf{w}_{ijt} + (\bar{z}_{ij} - \mu(z))\alpha_z\mathbf{w}_{ijt} + c_{ij} \tag{4}$$

Equation (2) states that conditioning on the heterogeneous coefficient (α_{ij}) and the bilateral fixed effects (c_{ij}), the vector z contributes no information about the expected (log) volume of trade. Equation (3) states that condition on the bilateral fixed effect, and all of the other standard gravity covariates, and z, the treatment effect depends on a constant and the deviation of the bilateral pair's vector time average of each z ($\bar{z}_{ij} = (\sum_{t=1}^{T_{ij}} z_{ijt})/T_{ij}$) from the world mean for $\mu(z)$.[1] By demeaning the vector of $z's$, α is the unconditional *average treatment effect* and the heterogeneity of the agreement is related to the observables ($(\bar{z}_{ij} - \mu(z))\alpha_z$).

To provide guidance for the variables included in z, we appeal to the literature on the "determinants of trade agreements." Recent empirical studies by economists and political scientists have identified several factors that are associated with the likelihood

[1]Since each bilateral pair may not have all observations for all periods in the data, we sum over the periods in which we do have observations and divide by the total number of observations for the bilateral pair (T_{ij}) as is standard we use the average across all bilateral pairs (N_{ij})as our measure of the mean; that is, ($\mu(z) = (\sum_{ij} \bar{z}_{ij})/N_{ij}$) (see Wooldridge, 2010).

of a country pair entering into an EIA. Using a model where the social planner's objective is to maximize the welfare for the representative (or average) citizen, Baier and Bergstrand (2004) show that geographic factors (distance, contiguity, and language, for example) are highly correlated with probability of country pair entering into a free trade agreement. In the context of the model, these factors increase welfare because they are associated with higher trade flows that associated with the market structure used in their model. Recent studies have extended their basic model to explain the spatial relationship among EIA's (Egger and Larch, 2008) and the evolution of trade agreements over time (Baldwin and Jaimovich, 2012); both find similar results as Baier and Bergstrand. Generally speaking, the factors that influence the likelihood of a trade agreement in the economic sphere are typically associated with higher welfare through higher trade flows. Thus, this literature suggests that these factors may also be correlated with a higher volume of trade; that is, there may be important interactions between EIAs and trade costs. Clearly, economists are not alone in examining the causes and consequences of economic integration agreements. Political scientists have investigated these issues as well. Researchers' interested in international relations and comparative politics have also identified factors that influence the likelihood of an agreement. In these studies, the political economy and market structure aspects may be important for the formation of a free trade agreement. For example, Grossman and Helpman (1995) show that economic integration agreements may occur if the agreement creates rents for the for particular interest groups even if the economy as a whole are made worse off. Milner (1997) was one of the first to argue that that exporters in industries that are characterized by economies of scale are more more

likely to press for a economic integration agreement. Aside from these political economy type of aspects, scholars in this area also identify factors related to *transactions costs* associated with negotiating and monitoring the agreements. Hurrell (1995) discusses the importance of domestic institutions and the viability of the institutions in the nation states for establishing economic integration across regions. Hurrell also discusses the importance of the integration agreements can be for "locking-in" domestic reforms may help promote more efficient domestic market reforms. Nogues and Quintanilla (1993) argue that integration agreements are more easily formed by states that have similar domestic institutions. As Mansfield and Solingen (2010) point out, states with highly dissimilar institutions have much to gain from forming a regional agreement, but the impediments to do so are so may be substantial. Similarly, states may form EIAs even though the change in trade flows from an agreement may not be as great as other potential agreements, but the costs of negotiation and monitoring may be significantly lower to make the agreement viable. It is possible these factors may not impact trade flows; rather they may increase the probability of trade agreements through lower negotiating and monitoring costs. In fact, it could be the case that if a bilateral pair has a common set of industrial regulations and laws, the EIA may not create as much trade as another bilateral pair that was similar on all other dimensions. Gruber (2000) makes a similar argument as Baier, Bergstrand and Marriutto (2013) that countries may enter into regional agreements even though gains from entering are not that great; however, the costs of being outside of the agreement may be too high. Mansfield and Solingen (2010) also argue that domestic security may be a driver that has impact if and when countries choose to enter into a trade

agreement.

The purpose of this paper is not to address all of these factors specifically. We choose a parsimonious set of controls to assess the extent to which these economic and political factors influence the heterogeneity of the agreements. In particular, our baseline specification is given by:

$$ln(TF_{ijt}) = \phi + \varphi^S_{it} + \varphi^D_{jt} + \alpha \mathrm{w}_{ijt} + \alpha^d(dist_{ij} - \bar{dist})\mathbf{w}_{ijt} + \alpha^A(adj_{ij} - \bar{adj})\mathrm{w}_{ijt} +$$

$$\alpha^L(lang_{ij} - \bar{lang})\mathrm{w}_{ijt} + \alpha^{CL}(legal_{ij} - \bar{legal})\mathrm{w}_{ijt} + \alpha^R(relig_{ij} - \bar{relig})\mathrm{w}_{ijt} + c_{ij} + \nu_{ijt} \quad (5)$$

where φ^S_{it} is a set of source-time fixed effects to account for any time-varying exporter effects, φ^D_{it} is a set of destination-time fixed effects to account for any time-varying importer effects, $dist_{ij}$ is the log of distance between countries i and j, adj_{ij} is a dummy variable that is equal to one if country i and country j share a common border, $lang_{ij}$ is a dummy variable if the two countries share a common language, $legal_{ij}$ is a dummy variable if the country pair has a common legal origin, $relig_{ij}$ is variable that measures the degree in which two countries share common religion. Let any variable $\bar{x}_{ij} = \sum_{\{ij\}}\left(\sum_{t=1}^{T_{\{ij\}}} x_{ijt}/T_{\{ij\}}\right)/N(ij)$ denote the mean of each variables. In this specification, other than the source and destination time effects, there are no time-varying bilateral effects.

The interaction terms contribute to the heterogeneity of the trade agreements. For example, suppose the coefficient on the interaction with distance is negative ($\alpha^D < 0$). This would imply that if a country pair had an EIA and their distance from each other was *less* than the average distance among all trading partners, the trade agreement would create more trade than would be the case if the distance from the each other were equal to the average. Because all of the *EIA* dummy interactions are demeaned, the average treatment effect of an economic integration agreement is equal to α in equation (5). This representation also allows us to compute different aspects of the treatment effect. For example, it is possible to calculate the average effect for only the pairs of countries that have a trade agreement (often referred to as the treatment on the treated). Formally, this can be expressed as:

$$ATT = \alpha + \alpha^d(\bar{dist}_{ij} - \bar{dist}) + \alpha^A(\bar{adj}_{ij} - \bar{adj}) + \alpha^L(\bar{lang}_{ij} - \bar{lang}) +$$

$$\alpha^{CL}(\bar{legal}_{ij} - \bar{legal}) + \alpha^R(\bar{relig}_{ij} - \bar{relig}) \tag{6}$$

where

- $\bar{dist}_{ij} = (\sum_{w_{ijt}=1} dist_{ij})/N(w_{ijt} = 1)$ is the mean distance for all countries that have an EIA,

- $\bar{adj}_{ij} = (\sum_{w_{ijt}=1} adj_{ij})/N(w_{ijt} = 1)$ is the fraction of countries that have an EIA that share a common border,

- $\bar{lang}_{ij} = (\sum_{w_{ijt}=1} lang_{ij})/N(w_{ijt} = 1)$ is the fraction of countries that have an EIA that share a common language,

- $\bar{legal}_{ij} = (\sum_{w_{ijt}=1} legal_{ij})/N(w_{ijt} = 1)$ is the fraction of countries that have an EIA that share a common legal origin, and

- $\bar{relig}_{ij} = (\sum_{w_{ijt}=1} relig_{ij})/N(w_{ijt} = 1)$ is the average religious association for the countries that have an EIA.

Similarly, we can define the average treatment on the untreated as:

$$ATU = \alpha + \alpha^d(\bar{dist}^C_{ij} - \bar{dist}) + \alpha^A(\bar{adj}^C_{ij} - \bar{adj}) + \alpha^L(\bar{lang}^C_{ij} - \bar{lang}) +$$

$$\alpha^{CL}(\bar{legal}^C_{ij} - \bar{legal}) + \alpha^R(\bar{relig}^C_{ij} - \bar{relig}) \tag{7}$$

where similar to above we define any variable $\bar{x}^C_{ij}$ as the average value for x among the bilateral pairs that do *not* have a economic integration agreement.

Once we control for unobserved heterogeneity that is potentially correlated with the trade flows, we can obtain consistent estimates of the average treatment effect (ATE),

the average treatment effect on the treated (ATT), and the average treatment effect on the untreated (ATU).

III. Heterogeneous Treatment Effects

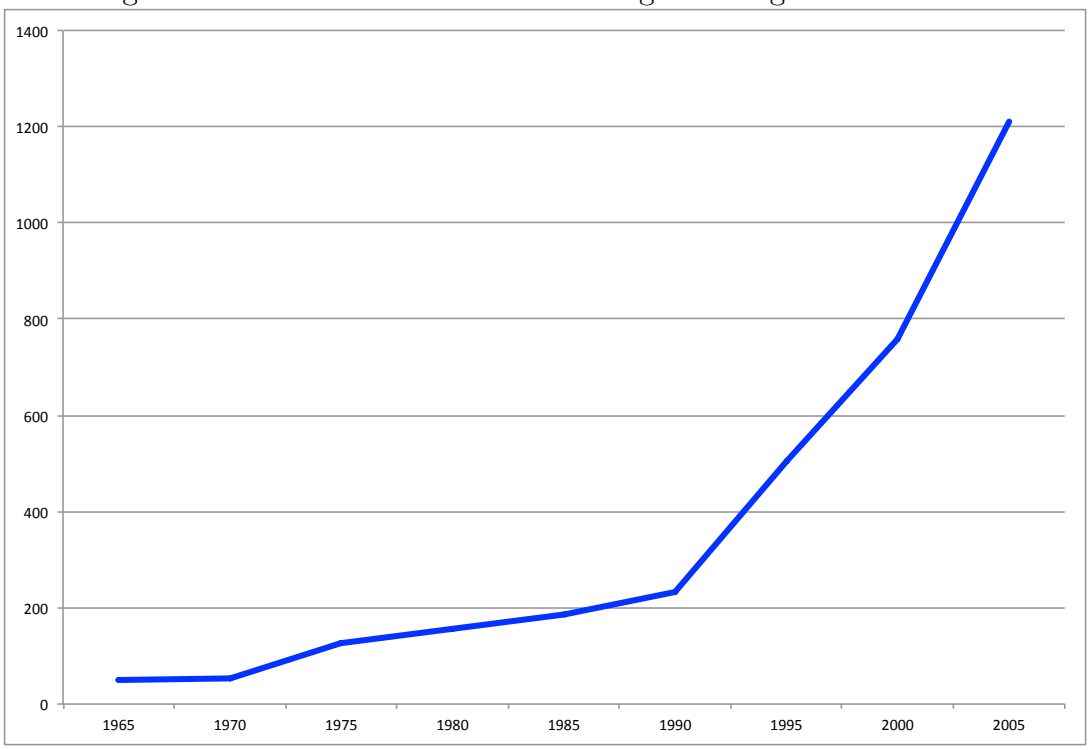
As is now common, we use panel techniques to remove the unobserved time-invariant heterogeneity. Our data consists of a panel (for every five years) from 1965 to 2005 of bilateral trade flows, bilateral distance, a set of dummy variables for adjacency, common language, religious similarity, common language, legal origin, and economic integration agreements among the 189 trading partners. Our baseline results are obtained using a linear fixed effects model. In Section A, we describe our data set. In section B, we provide estimates of the gravity equation assuming first a "homogeneous" treatment effect and then the "heterogeneous" treatment-effects model. In Section C, we examine the treatment effects on the treated and untreated. In D, we relate these results to the economic determinants of trade agreements.

III.A. Data

We describe briefly the data used in the estimation. Nominal trade flows are from the United Nation's COMTRADE database for the years 1965, 1970,..., 2005. Bilateral distances, language, adjacency, religion, and common legal origin are from the BACI data set. The data set for Economic Integration Agreements comes from Baier and Bergstrand data set (2013). There are 189 countries included in the data set. We

consider a bilateral pair in an EIA if the bilateral pair is in a free trade agreement, a customs union, a common market or an economic union. For the bilateral pairs where we observe trade flows and all of the covariates, Figure 1 depicts the rise in economic integration agreements over time.

Figure 1: The Growth of Economic Integration Agreements Over Time



In 1965, there were 50 bilateral pairs that had an economic integration agreements. By 1985, the amount of bilateral agreements more than tripled, and by the end of the sample there were more than 1200 bilateral arrangements![2]

[2]The increase in the number of economic integration agreements differs from the number in the introduction because we are conditioning on the set of countries where we have a complete set of gravity controls.

III.B. Empirical Results

Columns (1) and (2) of Table 1 provide empirical estimates using the linear fixed effects model. Column (1) provides standard estimate of the effect of trade agreements; that is, the heterogeneity is unobserved and uncorrelated with the other right-hand side variables. In this case, an EIA increases trade by roughly 50 percent ($exp(0.379) - 1 = 0.46$). When we allow for "observed" heterogeneity, the average treatment effect indicates that an EIA will increase trade flows by about 10% ($exp(0.0.96) - 1 = 0.10$). However, if a bilateral pair has an EIA and the distance between the pair is one standard deviation closer to each other than the world average, the effect of an EIA results in trade that is roughly 23 percentage points higher because of their proximity($exp(0.096 - 0.230 * (-0.825) - 1) = 0.331$). If the country pair has an EIA and the two share a common religion, all else held constant, the EIA will increase trade by nearly 45 percent ($exp(.096 + .406 * (1 - .297)) - 1 = 0.464$). If two countries share a common legal origin, this has a tendency to weaken the overall impact of an EIA. This finding may imply that a common legal system lowers the costs of negotiating a trade agreement which increases the likelihood of an agreement; however, because the common legal system may also imply more common regulations, the EIA may not result in "deeper integration" and increased trade that may occur when country pairs must *harmonize* regulations. Therefore, the likelihood of an economic integration agreement may increase even though trade flows may not. The relationship between legal origin and EIAs also stresses why we need to interpret these results with care. Taken on its own, a common legal origin would result in 10% lower trade ($exp(.096 - 0.332(1 - 0.343)) = 0.884$). However, countries that have an

Table 1: Empirical Estimates

	OLS		OLS $Hetergenous\ Treatment$	
EIA	0.379^{**} (0.0340)		0.0964^{+} (0.0527)	
$EIA * (\bar{dist}_{ij} - \bar{dist})$			-0.2301^{**} (0.0398)	
$EIA * (\bar{adj}_{ij} - \bar{adj})$			0.0044 (0.1087)	
$EIA * (\bar{lang}_{ij} - \bar{lang})$			0.0809 (0.0858)	
$EIA * (\bar{legal}_{ij} - \bar{legal})$			-0.3320^{**} (0.0677)	
$EIA * (\bar{relig}_{ij} - \bar{relig})$			0.4061^{**} (0.0834)	
N	120400		120400	
R^2	0.412		0.413	

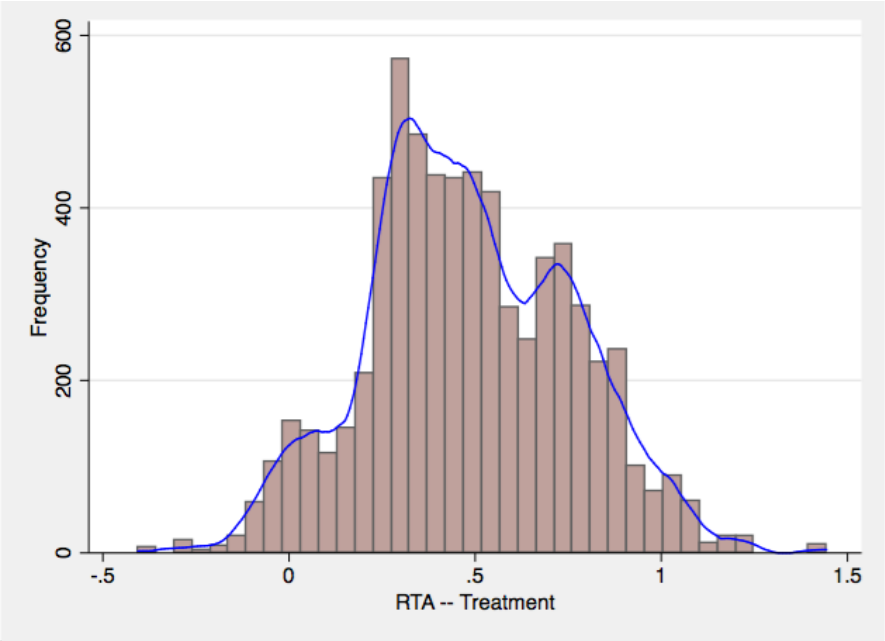
economic integration agreement and have the same legal origin also tend to be closer to each other and are more likely to share a common religion. Conditioning on these factors, trade is 30 percent higher when countries have the same legal origin and have an economic integration agreement.

III.C. Treatment on the Treated and Untreated

The example in the last section indicates the importance of conditioning on the controls. The random coefficient model introduces more heterogeneity into the measured impact of an economic integration agreement. Figure 2 depicts the frequency distribution across all years of average treatment effect for the country pairs that have an

EIA.

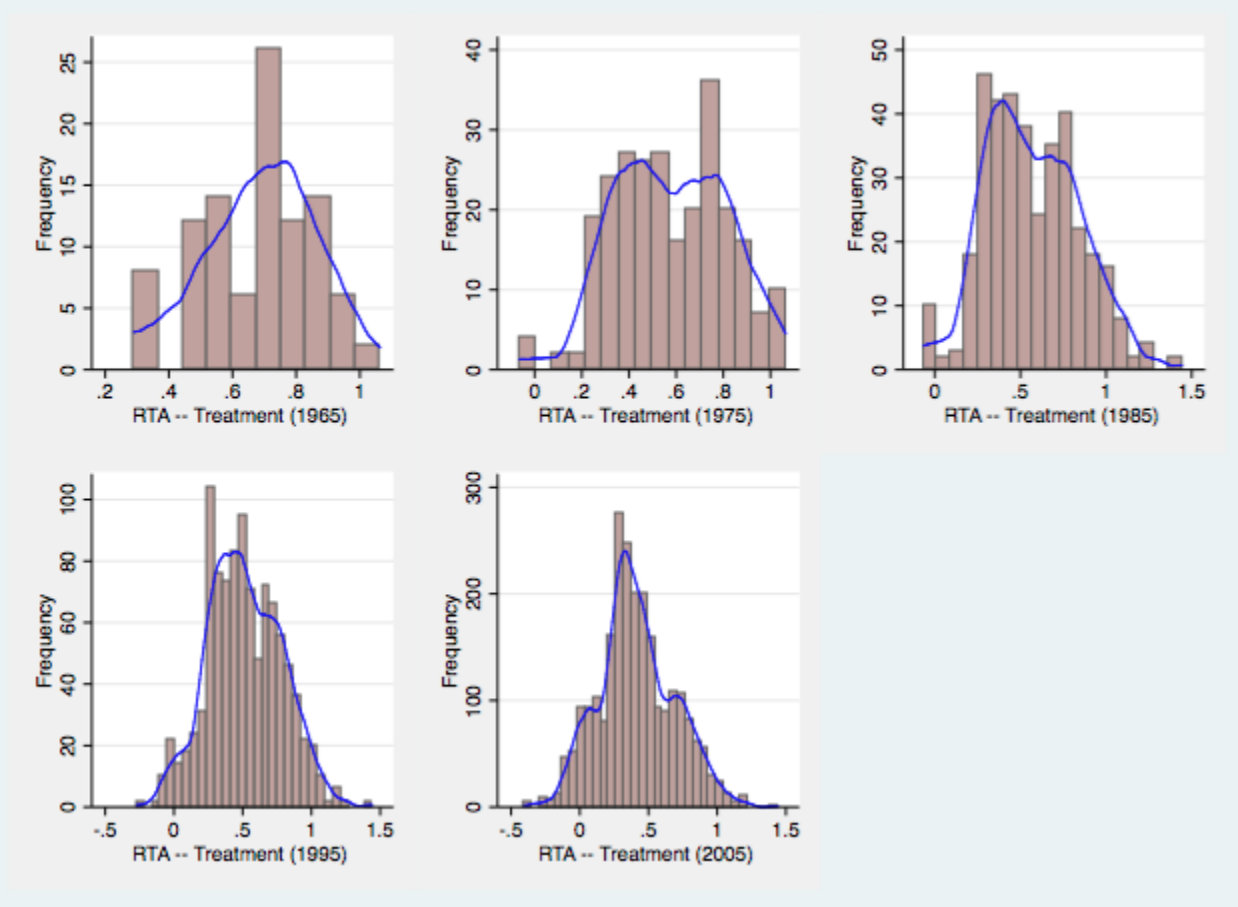
Figure 2: Impact of an EIA – All Country Pairs Over All Years



The kernel density plot indicates that averaged over all years the measured impact of trade agreements tends to increase trade by about 62 percent $(exp(.483) - 1)) = .62)$. While this is informative, it contains duplicate entries for countries that have had an agreement over several periods. Therefore, it may be more informative to look at the results for a given year. In 2005, among the bilateral pairs that had an EIA, the average impact of the economic integration agreement on trade flows was to increase bilateral trade by more than 50 percent $(exp(.415) - 1) = .51)$. The point estimates suggest of all the trade agreements more than 93 percent of the agreements resulted

in higher trade trade. For the 6.2% of the sample where the predicted effect was negative the EIA average decline in trade was nine percent. For the country pairs where the treatment effect is positive, trade tends to be about 55% higher. Figure 3 depicts the density plots for the years 1965, 1975, 1985, 1995 and 2005.

Figure 3: Treatment Effect on Treated: 1965-2005



It is interesting to note that in 1965 the typical EIA resulted in roughly a doubling

of trade ($exp(0.684) - 1 = .98$) and there were no agreements where the trade was predicted to be lower with the agreement than without. In 1985, the typical EIA increased trade by more than 75% ($exp(.569) - 1 = .766$), and by 2005 the average impact across all EIAs was slightly above 50 percent ($exp(.414) - 1 = .513$). These results imply that the original group of countries that entered into EIAs tended to, on average, be more trade creating than agreements that followed in subsequent years. The table below provides some additional insights.

Table 2: Trade Creation and Economic Integration Agreements

Year	Number of EIA's	ATT	Range	Number of ATT < 0	ATU
1965	100	0.684	(0.29, 1.06)	0	0.062
1975	256	0.579	(-0.06, 1.07)	4	0.032
1985	373	0.570	(-0.07, 1.44)	6	0.024
1995	1013	0.516	(-0.27, 1.44)	30	0.020
2005	2419	0.414	(-0.41, 1.44)	152	-0.002

Column 3 reports the point estimate of the fixed effects regression so that the effect on trade can be obtained by taking the exponential of this value. The table indicates that over time the average treatment effect on the treated declines over time, but the decline is not monotonic. Column 4 indicates that the range of the estimates has increased over time. Also, the number of agreements where the empirical estimates of the treatment effect was less than zero increased over time. The final column reports average treatment effects on the untreated. The average treatment effect on the untreated declines over time as well. Since the average treatment effect on the treated and the average treatment effect on the untreated decline over time, this suggests that over time, on average, the country pairs that sign economic integration

agreements earlier are the ones that tend to be the agreements that are relatively more trade creating.

It is also instructive to look at a few agreements in particular. In 2005, the average treatment effect for the countries in the European Union is about 73 percent ($exp(.545) - 1$). Belgium and Germany have the largest treatment effect 1.89 ($exp(1.06) - 1$) and there are only two bilateral pairs where the expected trade flows are lower because of the EIA (Greece-Spain and Greece-Portugal). For NAFTA, the average treatment effect is nearly 60 percent ($exp(0.456) - 1$). Among the countries in MERCUSOR, the average treatment effect of the economic integration agreement is nearly 60 percent as well. The random coefficient model predicts that largest ATT is for Argentina and Uruguay (average trade roughly doubles); while the least amount of trade is created between Uruguay and Venezuela (20 percent higher). Finally, another way to break the data down is to report the ATT for different levels of integration. In 2006, the ATT for the countries that have a free trade agreement is 41%, for the countries that are in a customs union the average treatment effect is 51 percent, for those in a common market the ATT is 70 percent, and those in an economic union the ATT is 66 percent. Overall, the random coefficient model is consistent with more trade being created the deeper the integration and is consistent with bilateral pairs "choosing wisely"; that is, the bilateral pairs that integrate over time tend to, on average, be the ones the have the most to gain in terms of more bilateral trade.

Conclusion

The objective of this paper was to link to the causes and consequences of regionalism and show how these two strands of literature are informative about each other. We show that by incorporating the insights from the *determinants of economic integration* into the models of bilateral trade flows, we can obtain heterogeneous effects of trade agreements. These effects seem to be consistent with the evolution of trade agreements over time – countries choose wisely, and the results appear to be consistent with our priors on the trade creating aspects of economic integration agreements.

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